STAT2003/STAT7003
Material Excluded from Examination

Semester 1, 2026

The following lists the sections, claims, and examples from the course notes that are **not examinable**. Everything in the notes not listed here *is* examinable. We still recommend reading and following all of the material, as it provides context and deeper understanding.

Chapter 1: Random Experiments, Probability Spaces, and Simulation

1.1 Random Experiments, Reality, and Simulation

- Example: Machine Lifetime

1.3 The Probability Triple, Axioms, and Properties

- Subsection: Sigma-Algebras
- Subsection: Continuity of Probability

1.4 More on Simulation

- Subsection: Pseudorandom Sequence Algorithms — specifically, the details of Linear Congruential Generators

Chapter 2: Counting, Inclusion-Exclusion, and Independence

2.3 Inclusion-Exclusion Principle

- Proof of Claim 4 (Inclusion-Exclusion Principle for n events)

Chapter 3: Discrete Random Variables, their Distributions, and Moments

3.1 Discrete Random Variables and Their Distributions

- Proof of Claim 7 (Right-Continuity of the CDF)

3.2 Moments: Expectation, Mean, and Variance

- Proof (explanation) of Claim 8 (LOTUS — Law of the Unconscious Statistician)
- Summation formulas (3.50)–(3.52) — these will be provided if needed

3.4 Bernoulli Trials and Related Distributions

- Direct calculation of the mean of the Binomial distribution (equations (3.78)–(3.81))

3.5 Poisson and Hypergeometric Distributions

- Verification that the Hypergeometric pmf sums to 1 (the Vandermonde identity, equation (3.106))
- Proof of Claim 16 (Hypergeometric–Binomial Convergence)

Chapter 4: Conditional Probability and Markov Chains

4.4 Markov Chains Introduction

- Subsection: Countably Infinite State Spaces

4.6 A Peek at Markov Chain Properties

- All of this section *except* the subsection on Limiting Distributions

Chapter 5: Continuous Random Variables and their Distributions

5.2 Expectation, Moments, and Properties

- Skewness and Kurtosis (including the Example on $f(x) = 2x$)

5.5 The Gamma and Beta Distributions

- Reparametrisation by Mean and Coefficient of Variation
- Transformations of the Beta Distribution

Chapter 6: More Aspects of Random Variables and Distributions

6.1 Transformations of Random Variables

- Proof of the Change-of-Variable Formula claim

6.2 The Inverse Probability Transform

- The Quantile Function and the General Case (i.e. the quantile function for general, non-continuous distributions)

6.3 The Markov and Chebyshev Inequalities

- Further Inequalities

6.4 Heavy Tails, Infinite Moments, and Undefined Means

- Proof of the Moment Hierarchy claim

Chapter 7: Joint Distributions

7.3 Marginal and Conditional Distributions

- The precise argument for how the limiting conditional cdf $F_Y(y \mid x < X \leq x+h)$ converges, as $h \to 0$, to the conditional cdf $F_{Y|X}(y \mid x)$. The booklet gives only the rough idea (approximating each integral by $h\, f(\cdot)$ plus higher-order terms); a rigorous derivation of this limit is not examinable.

7.6 The Multivariate Normal Distribution

- Proof of the Affine Combinations of Normals claim

Chapter 8: More Aspects of Joint Distributions

8.5 Generating Multivariate Normal Distributions

- The Cholesky decomposition is not examinable: students are not expected to compute a Cholesky factor by hand, nor to know the theory behind why a positive-definite matrix admits one.

Chapter 9: Moment Generating Functions, Related Transforms, and Examples

9.2 Other Transforms: Laplace–Stieltjes and Characteristic Functions

- The characteristic function. It may be discussed briefly in class for context, but is not examinable; no familiarity with complex numbers is required.

Chapter 10: Random Samples and Limit Theorems

10.2 Properties of Normal Samples

- The t-distribution and the F-distribution, and their relationship to normal samples (the corresponding test statistics built from the sample mean and sample variance).